

Sai Rahul Bhat Gaadhi Raju

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LinkedIn — GitHub

Career Objective

B.Tech ECE student interested in semiconductor technology, VLSI design, and fabrication. Seeking an internship to apply skills in digital design, signal processing, and system-level design.

Education

B.Tech in Electronics and Communication Engineering	2024 – Till Date
Anurag University, Hyderabad CGPA: 8.01	
Diploma in Electronics and Communication Engineering	2021 – 2024
JNGPT, Ramanthapur CGPA: 9.2	

Technical Skills

HDL & Programming: Verilog HDL, C (Basics), Python
Tools: Vivado, MATLAB, Cadence Virtuoso Tools
Core: Digital Electronics, VLSI, RTL Design, Semiconductor Fundamentals, Signal Processing

Projects

Pipelined ALU Design (RTL – Verilog)

- Designed multi-stage pipelined ALU based on RISC-V to improve throughput
- Developed pipeline stages considering timing constraints
- Verified functionality using Vivado

Barrel Shifter Design (Verilog HDL)

- Designed N-bit barrel shifter using combinational logic
- Implemented efficient bit shifting using RTL modeling
- Verified design using simulation tools

4-bit ALU Design (Verilog HDL)

- Designed arithmetic and logical operations using modular RTL design
- Verified correctness using Vivado

VLSI Voyage: 16-bit ALU (Pipeline vs Non-Pipeline)

- Compared pipelined and non-pipelined ALU architectures
- Analyzed throughput, latency, and clock improvements
- Studied trade-offs between performance and hardware complexity

Certifications

- NPTEL – Digital Design with Verilog (IIT Guwahati)
- NPTEL – VLSI Design Flow RTL to GDS (IIIT Delhi)
- Maven Silicon – VLSI SoC Design Overview

Internship

Python Intern – Vault of Codes	1 Month
• Worked on Python fundamentals and logic-based problem solving	

Languages

English, Hindi, Telugu